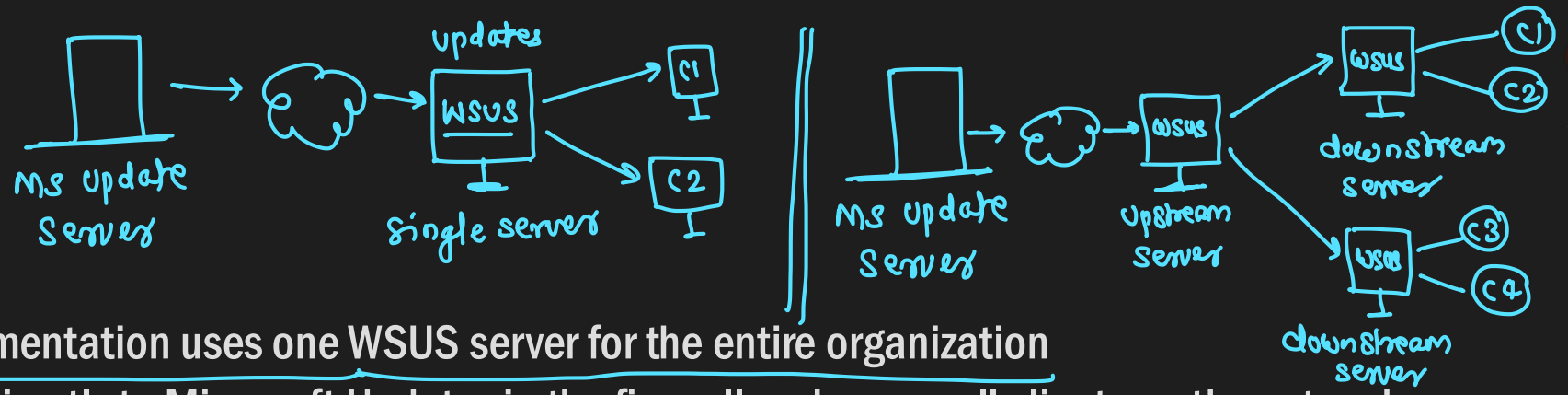
WSUS

Update Services

Introduction

- Windows Server Update Services (WSUS) is a server role that enables administrators to centrally manage and distribute Microsoft updates to computers across their network
- It acts as a middleman between Microsoft Update and your client machines, allowing you to test, approve, and deploy patches on your own schedule
- **Hardware Requirements**
  - **Processor:** 1.4 GHz x64 (2 GHz+ recommended)
  - **Memory:** Additional 2 GB RAM for WSUS (beyond the OS needs)
  - **Disk Space:** 40 GB+ recommended
- **Software Prerequisites**
  - **Updates:**
    - Install the February 2023 cumulative update (or newer) on WSUS servers running Server 2016/2019/2022
  - **Database:**
    - WSUS requires a database
    - It installs **Windows Internal Database (WID)** by default, or you can use SQL Server (Standard, Enterprise, or Express)

Deployment Models



- **Core Deployment Models**
  - **Single Server**
    - The most basic implementation uses one WSUS server for the entire organization
    - This server connects directly to Microsoft Update via the firewall and serves all clients on the network
    - This model is typically used in small networks with a single physical location and is the simplest to manage
  - **Multiple Independent Servers**
    - If your organization has multiple isolated physical locations, you might deploy a WSUS server at each site.
    - Each server maintains its own Internet connection to download updates directly from Microsoft Update
    - This model requires more administrative effort because each server must be managed independently—downloading, approving, and managing client targeting separately for each location
- **Hierarchical Deployment Models**
  - For larger organizations, WSUS servers can be organized in a hierarchy where downstream servers synchronize from an upstream server rather than directly from Microsoft Update
  - This approach reduces Internet bandwidth consumption by downloading updates once and then distributing them internally
  - **Important constraint**: Microsoft recommends a maximum hierarchy depth of **three levels** for WSUS servers

WID vs SQL Server

Feature	Windows Internal Database (WID)	SQL Server (Full/Express)
License Cost	Included with Windows Server	Requires separate license (Full edition)
Performance (Single Server)	Same as SQL Server	Same as WID
NLB / High Availability	**Not supported**	**Required** for NLB clusters
Database Size Limit	No built-in limit	Express edition: **10 GB** limit
Remote Management	Not possible remotely	Full remote management via SSMS
Authentication	Windows Authentication only	Windows Authentication only (WSUS requirement)

When to Choose Which

- **Choose WID when**
  - You are deploying a **single WSUS server** (or multiple independent servers)
  - You have **no existing SQL Server infrastructure** and want to avoid licensing costs
  - You do **not** need Network Load Balancing (NLB)
  - You are deploying WSUS in **branch offices** with separate databases per server
- **Choose SQL Server when**
  - You need an **NLB WSUS solution** for high availability or large-scale client load
  - You **already have SQL Server installed** and want to consolidate database management
  - You need **remote database management** capabilities

Configuration

- **Step 1: Install the WSUS Role**
  - In **Server Manager**, use "Add Roles and Features" to install the **Windows Server Update Services** role
  - During the post-installation configuration wizard, Choose to synchronize from **Microsoft Update**
  - Select the **Languages**, **Products** (e.g., Windows, Office), and **Classifications** (e.g., Critical, Security) you want to download
- **Step 2: Configure Client Computers via GPO**
  - This is the critical step to point your clients to the WSUS server
  - Create a GPO linked to the OU containing your computers, and configure these settings under
    - Computer Configuration > Policies > Administrative Templates > Windows Components > Windows Update
  - **Specify intranet Microsoft update service location**
    - Set both URLs to your WSUS server (http://wsus-server:8530)
  - **Configure Automatic Updates**: Set to **Enabled**
- **Step 3: Approve and Deploy Updates**
  - In the **WSUS Administration Console**, you approve updates for specific computer groups:
  - Go to **Updates** and filter by "Updates needed by computers".
  - Select the updates for your test group, right-click, and select **Approve**.
  - Set the approval to **"Approved for Install"**

Post Installation

- **Configure the Firewall**
  - Ensure client computers can reach the WSUS server.
  - The default ports are **8530 (HTTP)** and **8531 (HTTPS)**
  - The WSUS server itself must also be allowed outbound access to Microsoft domains on ports **80 and 443**
- **Set Up Computer Groups**
  - Create groups in the WSUS console to stage update deployments
  - You can assign clients to these groups manually or use **Client-Side Targeting** via Group Policy
- **Configure Client Machines**
  - Use Group Policy to point your domain computers to the WSUS server
  - Create a GPO with the policy **"Specify intranet Microsoft update service location"** under
    - Computer Configuration > Administrative Templates > Windows Components > Windows Update
- **Approve Updates**
  - After the first synchronization, updates will be listed in the console but not yet approved for installation
  - You must right-click the updates you want to deploy, select **Approve**, and choose the target computer group

Operational Best Practices



- Capacity and Architecture: Avoid the Single Server Trap
- Essential IIS Tuning to Prevent Crashes
- Security: Protect the Connection
- The Golden Rule of Maintenance: Decline Superseded Updates
- Client Targeting: Use Group Policy for Scale
- Sync Only What You Need
- Use SQL Server for large organization
- Monitor reports regularly
- Backup of your update database regularly